

Adding Swap Partitions

Using swap on Linux is a convenient way to improve Linux kernel memory usage. If a shortage of physical RAM occurs, non-recently-used memory pages can be moved to swap, which makes more RAM available for programs that need access to memory pages.

1. Use `fdisk /dev/sdb` to open your disk in `fdisk`
2. Press `n` to add a new partition. Specify start and stop cylinders and size.
3. Type `t` to change the partition type. If you are using `fdisk`, use partition type 82.
If you are using `gdisk`, use partition type 8200.
4. Use `mkswap` to format the partition as swap space. Use, for instance, `mkswap /dev/sdb1` if the partition you have just created is `/dev/sdb1`.
5. Type `free -m`. You see the amount of swap space that is currently allocated.
6. Use `swapon` to switch on the newly allocated swap space. If, for instance, the swap device you have just created is `/dev/vda6`, use `swapon /dev/vda6` to activate the swap space.
7. Type `free -m` again. You see that the new swap space has been added to your server.

Mounting File Systems

Mounting process is to make file system accessible through creating mount point

In windows mounting process is under letters like (c: - d: - e:etc) but in linux you can mount any partition or file system under directory (folder) so you have to create mount point (`mkdir /mount`).

Types of mounting :

- Temporary (in running time and still take effect as long as system up and running but when you reboot the system everthing is gone.
- Permanent through adding entries in `/etc/fstab` file.

To mount a file system, some information is needed:

- **What to mount:** This information is mandatory and specifies the name of the device that needs to be mounted.
- **Where to mount it:** This is also mandatory information which specifies the directory on which the device should be mounted.
- **What file system to mount:** Optionally, you can specify the file system type. In most cases, this is not necessary. The **mount** command will detect which file system is used on the device and make sure the correct driver is used.
- **Mount options:** Many mount options can be used when mounting a device. Using options is optional and depends on the needs you may have with the file system.

Manually Mounting File Systems

Mounting >>> `mount /dev/vda5 /mount`

unmounting >>> `umount /dev/vda5` or `umount /mount`

You can mount the partition through :

1. Device name
2. Label
3. UUID (best way to mount)

To make Label to a partition :

For ext4 file system

```
#e2label /dev/sdb1 data
```

```
#mount LABEL=data /mount
```

For xfs file system

```
#xfs_admin -L sales /dev/sdb1
```

```
#mount LABEL=sales /dev/sdb1
```